Lecture 1 – Mathematical Preliminaries

COSE215: Theory of Computation

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PLRG

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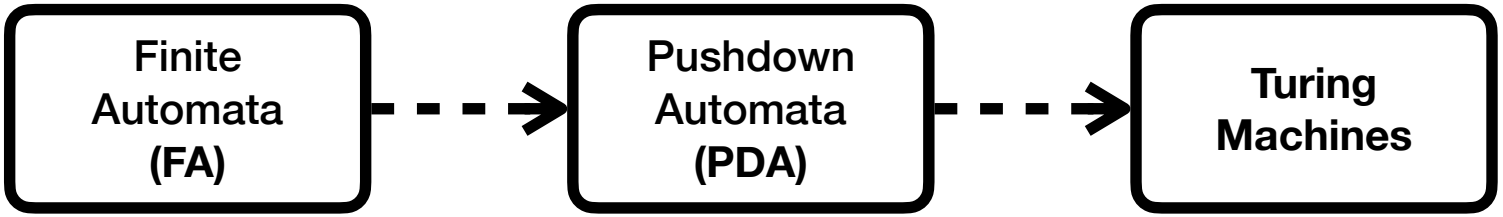
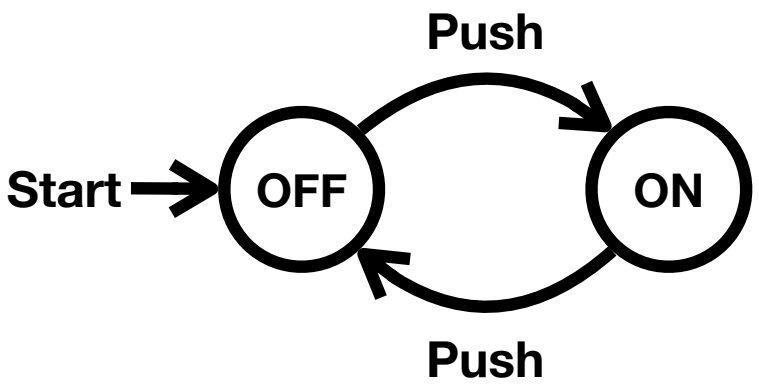
Roadmap: Towards Turing Machine

A Turing machine is a specific kind of **automaton**.



- **Part 1: Finite Automata (FA)**
  - Regular Expressions (REs)
  - Regular Languages (RLs)
  - Applications: text search, etc.
- **Part 2: Pushdown Automata (PDA)**
  - Context-Free Grammars (CFGs)
  - Context-Free Languages (CFLs)
  - Applications: programming languages, natural language processing, etc.
- **Part 3: Turing Machines (TMs)**
  - Lambda Calculus (LC)
  - Recursively Enumerable Languages (RELs)
  - Undecidability and Intractability

Recall



Theorem

The current state is OFF *if and only if the button is pushed* **even** *times.*

- Is it possible to prove it?

Let's learn **mathematical background and notation.**

Contents

1. Mathematical Notations
   - Notations in Logics
   - Notations in Set Theory

2. Inductive Proofs
   - Inductions on Integers
   - Structural Inductions
   - Mutual Inductions

3. Notations in Languages
   - Symbols & Words
   - Languages

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Notations in Logics

Notation	Description
A, B	arbitrary **statements**.
$P(x)$	a **predicate** is a statement having variables (e.g., x).
$A \wedge B$	the **conjunction** of A and B. (i.e., "A and B").
$A \vee B$	the **disjunction** of A and B. (i.e., "A or B").
$\neg A$	the **negation** of A. (i.e., "not A").

$$(\textbf{De Morgan's Laws}) = \begin{cases} \neg(A \wedge B) = \neg A \vee \neg B \\ \neg(A \vee B) = \neg A \wedge \neg B \end{cases}$$

$(\textbf{Truth Table}) =$

A	B	$A \wedge B$	$\neg(A \wedge B)$	$\neg A$	$\neg B$	$\neg A \vee \neg B$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Notations in Logics

Notation	Description
$A \Rightarrow B$	the **implication** of A and B (i.e., "if A then B" or "A implies B") (i.e., $\neg A \vee B$).
$A \Leftrightarrow B$	A **if and only if (iff)** B (i.e., $A \Rightarrow B \wedge B \Rightarrow A$).
$\forall x \in X.\ P(x)$	the **universal quantifier** (i.e., "for all x in X, $P(x)$ holds").
$\exists x \in X.\ P(x)$	the **existential quantifier** (i.e., "there exists x in X such that $P(x)$ holds").

(**Truth Table**) =

A	B	$A \Rightarrow B$	$\neg A$	B	$\neg A \vee B$
T	T	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	F	T

Notations in Set Theory

- A **set** is a collection of elements. For example,
  - (Integers) $= \mathbb{Z} = \{\cdots, -2, -1, 0, 1, 2, \cdots\}$
  - (Natural Numbers) $= \mathbb{N} = \{0, 1, 2, \cdots\}$
  - (Squares of $\mathbb{N}$) $= \{x^2 \mid x \in \mathbb{N}\} = \{0, 1, 4, 9, 16, 25, 36, \cdots\}$
  - (Even Numbers) $= \{x \in \mathbb{N} \mid x \equiv 0 \pmod 2\} = \{0, 2, 4, 6, 8, \cdots\}$

$$\textit{where} \qquad a \equiv b \pmod n \iff \exists k \in \mathbb{Z}.\ a = b + kn$$

  (i.e., a is **congruent** to b **modulo** n)
- The **empty set** is denoted by $\varnothing$.
- The **cardinality** of a set X is denoted by $|X|$.
- x is an **element** of a set X is denoted by $x \in X$.
- X is a **subset** of Y is denoted by $X \subseteq Y$.

$$X \subseteq Y \iff \forall x \in X.\ x \in Y$$

- X is a **proper subset** of Y is denoted by $X \subset Y$.

$$X \subset Y \iff X \subseteq Y \wedge X \neq Y$$

Notations in Set Theory

- The **union** of sets

$$\begin{aligned} X \cup Y &= \{x \mid x \in X \vee x \in Y\} \\ \bigcup \mathcal{C} &= X_1 \cup X_2 \cup \cdots \cup X_n = \{x \mid \exists X \in \mathcal{C}.\ x \in X\} \end{aligned}$$

where $\mathcal{C} = \{X_1, X_2, \cdots, X_n\}$.

- The **intersection** of sets

$$\begin{aligned} X \cap Y &= \{x \mid x \in X \wedge x \in Y\} \\ \bigcap \mathcal{C} &= X_1 \cap X_2 \cap \cdots \cap X_n = \{x \mid \forall X \in \mathcal{C}.\ x \in X\} \end{aligned}$$

where $\mathcal{C} = \{X_1, X_2, \cdots, X_n\}$.

- The **difference** of sets

$$X \setminus Y = \{x \mid x \in X \wedge x \notin Y\}$$

Notations in Set Theory

- The **complement** of a set X is denoted by $\overline{X}$.

$$\overline{X} = \{x \mid x \in U \wedge x \notin X\}$$

where U is the **universal set**.

- The **power set** of a set X is denoted by 2^X or $\mathcal{P}(X)$.

$$2^X = \mathcal{P}(X) = \{Y \mid Y \subseteq X\}$$

- The **Cartesian product** of sets X and Y is denoted by $X \times Y$.

$$X \times Y = \{(x, y) \mid x \in X \wedge y \in Y\}$$

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    Structural Inductions
    Mutual Inductions

3. Notations in Languages
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Inductions on Integers

Definition (Inductions on Integers)

Let $P(n)$ be a predicate on integers, and if

- **(Basis Case)** $P(k)$ holds where k is an integer, and
- **(Induction Case)** for all integer $n \geq k$, $P(n) \Rightarrow P(n+1)$,

then $P(i)$ holds for all $i \geq k$.

$P(n)$ is called **induction hypothesis** (I.H.).

$$\underset{\text{Basis}}{P(k)} \xrightarrow{\text{Induction}} P(k+1) \xrightarrow{\text{Induction}} \cdots \xrightarrow{\text{Induction}} P(n) \xrightarrow{\text{Induction}} P(n+1)$$

Inductions on Integers – Example 1

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Example

Prove that $\forall n \geq 0.\ \sum_{i=0}^{n} i = \frac{n(n+1)}{2}$.

Proof)

- (Basis Case): $0 = 0(0+1)/2$ $\square$
- (Induction Case): Assume that it holds for n (I.H.)

  Then, let's prove it for $n+1$:

$$\begin{aligned}\sum_{i=0}^{n+1} i &= (n+1) + \sum_{i=0}^{n} i \\ &= (n+1) + \frac{n(n+1)}{2} \qquad (\because I.H.) \\ &= \frac{(n+1)(n+2)}{2} \quad \square\end{aligned}$$

Inductions on Integers – Example 2

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Example

Prove that $\forall n \geq 0.\ \sum_{i=0}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$.

Proof)

- (Basis Case): $0^2 = 0(0+1)(2*0+1)/6$ $\square$
- (Induction Case): Assume that it holds for n (I.H.).

  Then, let's prove it for $n+1$:

$$
\begin{aligned}
\sum_{i=0}^{n+1} i^2 &= (n+1)^2 + \sum_{i=0}^{n} i^2 \\
&= (n+1)^2 + \frac{n(n+1)(2n+1)}{6} \qquad (\because I.H.) \\
&= \frac{(n+1)(n+2)(2(n+1)+1)}{6} \qquad \square
\end{aligned}
$$

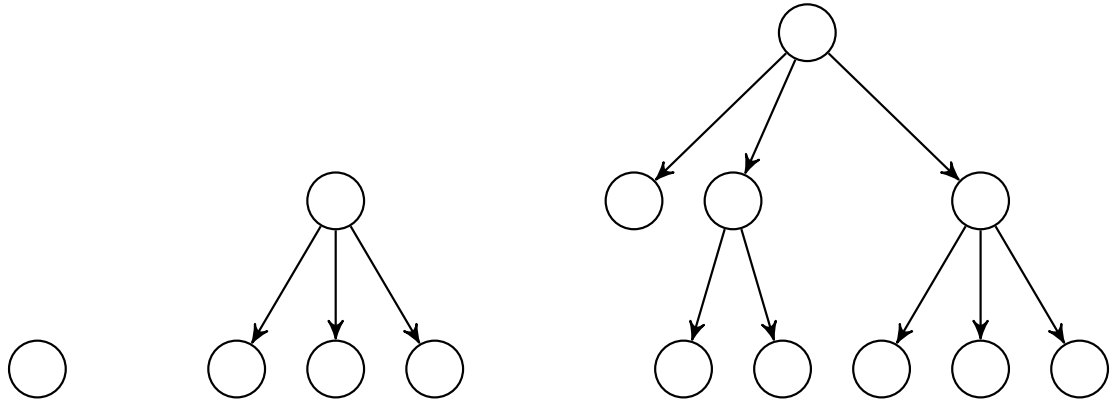
Structural Inductions – Inductive Definitions

In CS, we often define somethings as **inductively-defined sets**.
For example, we can define **trees** as follows:

Example (Inductive Definition of Trees)

A **tree** is defined as follows:

- **(Basis Case)** A single **node** N is a tree.
- **(Induction Case)** If $T_1, \cdots, T_n$ are trees, then a graph defined with a new root node N and edges from N to $T_1, \cdots, T_n$ is a tree.



Structural Inductions – Inductive Definitions

Another example is a set of **arithmetic expressions**:

Example (Inductive Definition of Arithmetic Expressions)

An **arithmetic expression** is defined as follows:

- **(Basis Case)** A **number** or a **variable** is an arithmetic expression.
- **(Induction Case)** If E and F are arithmetic expressions, then so are $E{+}F$, $E{*}F$, and (E).

`42`	`x`	`x + y`
`42 * x`	`(x)`	`(x * y) * z`
`(2 + x) * y`	`x * (x * y)`	`((((x))))`

Structural Inductions

Definition (Structural Inductions)

Let $P(x)$ be a predicate on a **inductively-defined set** X, and if

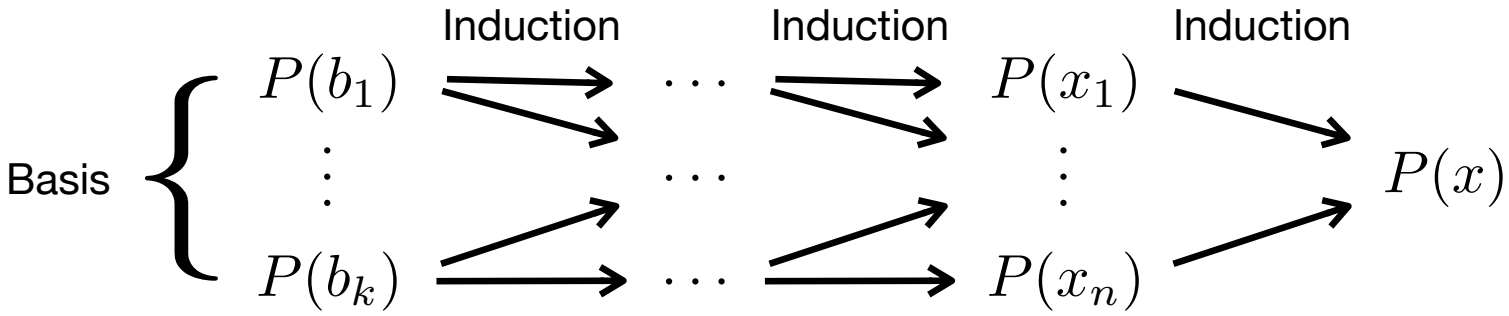
- **(Basis Case)** $P(b_1), \cdots, P(b_k)$ hold for all basis cases $b_1, \cdots, b_k$.
- **(Induction Case)** for all $x \in X$,

$$P(x_1) \wedge \cdots \wedge P(x_n) \Rightarrow P(x)$$

  where $x_1, \cdots, x_n$ are the **sub-structures** of x.

then $P(x)$ holds for all $x \in X$.

$P(x_1), \cdots, P(x_n)$ are called **induction hypotheses**.



Structural Inductions – Example 1

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Example

Prove that for all tree T, the number of nodes in T is equal to the number of edges in T plus one.

Proof) Let $N(T)$ be the number of node and $E(T)$ be the number of edges in T. Let's prove $\forall T.\ N(T) = E(T) + 1$.

- (Basis Case): $N(T) = 1$ and $E(T) = 0$. $\square$
- (Induction Case): Assume that it holds for $T_1, \cdots, T_n$ (I.H.). Then,

$$
\begin{aligned}
N(T) &= 1 + \sum_{i=1}^{n} N(T_i) \\
&= 1 + \sum_{i=1}^{n} (E(T_i) + 1) \qquad (\because I.H.) \\
&= 1 + n + \sum_{i=1}^{n} E(T_i) \\
&= 1 + E(T) \quad \square
\end{aligned}
$$

Structural Inductions – Example 2

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Example

Prove that for all arithmetic expression E, the number of left parentheses in E is equal to the number of right parentheses in E.

Proof) Let $L(E)$ be the number of left parentheses and $R(E)$ be the number of right parentheses in E. Let's prove $\forall E.\ L(E) = R(E)$.

- (Basis Case): $L(E) = R(E) = 0$ for numbers and variables. $\square$
- (Induction Case): Assume that it holds for E and F (I.H.). Then,

$$
\begin{aligned}
L(E{+}F) &= L(E) + L(F) = R(E) + R(F) \qquad (\because I.H.) \\
&= R(E{+}F) \quad \square \\
L(E{*}F) &= L(E) + L(F) = R(E) + R(F) \qquad (\because I.H.) \\
&= R(E{*}F) \quad \square \\
L((E)) &= L(E) + 1 = R(E) + 1 \qquad (\because I.H.) \\
&= R((E)) \quad \square
\end{aligned}
$$

Mutual Inductions

Sometimes, we need to prove multiple predicates simultaneously.

Definition (Mutual Inductions)

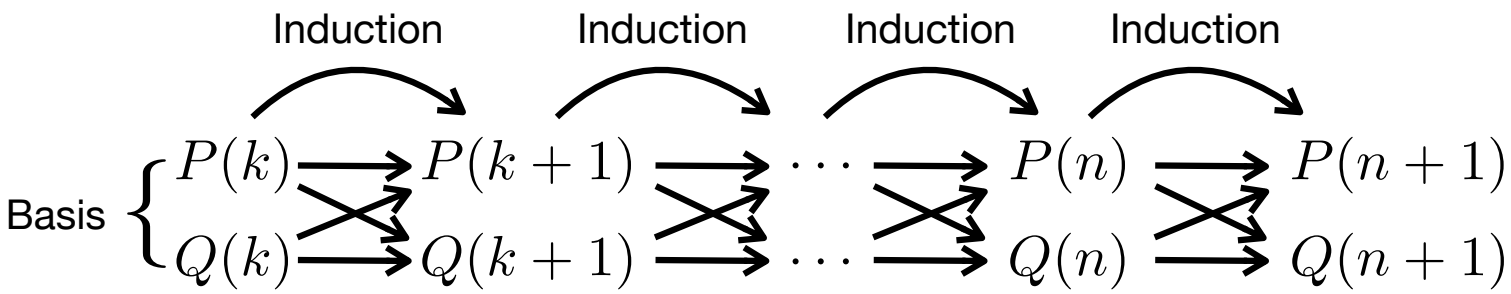
Let $P(x)$ and $Q(x)$ are predicates on integers, and if

- **(Basis Case)** $P(k)$ and $Q(k)$ hold where k is an integer, and
- **(Induction Case)** for all $n \geq k$,

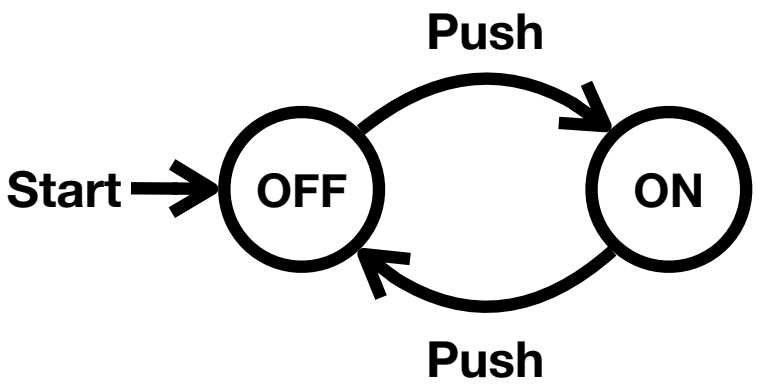
$$P(n) \wedge Q(n) \Rightarrow P(n+1) \wedge Q(n+1)$$

then $P(i)$ and $Q(i)$ hold for all $i \geq k$.

$P(n)$ and $Q(n)$ are called **induction hypotheses**.



Mutual Inductions – Example



Theorem

The current state is OFF *if and only if the button is pushed* **even** *times.*

It is difficult to prove it with only one statement.

Let's prove it with two statements:

Theorem

The current state is OFF *if and only if the button is pushed* **even** *times, and the current state is* ON *if and only if the button is pushed* **odd** *times.*

Mutual Inductions – Example

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Theorem

The current state is OFF *if and only if the button is pushed* **even** *times, and the current state is* ON *if and only if the button is pushed* **odd** *times.*

Proof) Let $S(i)$ be the current state after i times of pushing. Let's prove

$$\forall i \geq 0.\ S(i) = \text{OFF} \iff i \equiv 0 \pmod 2 \qquad (P)$$
$$\forall i \geq 0.\ S(i) = \text{ON} \iff i \equiv 1 \pmod 2 \qquad (Q)$$

- <u>(Basis Case)</u>: Known facts: $S(0) = \text{OFF}$ and $0 \equiv 0 \pmod 2$
  - <u>$(P, \Rightarrow)$</u>: $S(0) = \text{OFF} \Rightarrow 0 \equiv 0 \pmod 2$ because $0 \equiv 0 \pmod 2$
  - <u>$(P, \Leftarrow)$</u>: $S(0) = \text{OFF} \Leftarrow 0 \equiv 0 \pmod 2$ because $S(0) = \text{OFF}$
  - <u>$(Q, \Rightarrow)$</u>: $S(0) = \text{ON} \Rightarrow 0 \equiv 1 \pmod 2$ because $S(0) \neq \text{ON}$
  - <u>$(Q, \Leftarrow)$</u>: $S(0) = \text{ON} \Leftarrow 0 \equiv 1 \pmod 2$ because $0 \not\equiv 1 \pmod 2$

Mutual Inductions – Example

- <u>(Induction Case)</u>: Assume that it holds for n (I.H.):

$$S(n) = \text{OFF} \iff n \equiv 0 \pmod 2 \qquad (P - I.H.)$$
$$S(n) = \text{ON} \iff n \equiv 1 \pmod 2 \qquad (Q - I.H.)$$

  - <u>$(P, \Leftrightarrow)$</u>:

$$\begin{aligned} S(n+1) = \text{OFF} &\iff S(n) = \text{ON} \\ &\iff n \equiv 1 \pmod 2 \qquad (\because Q - I.H.) \\ &\iff n + 1 \equiv 0 \pmod 2 \end{aligned}$$

  - <u>$(Q, \Leftrightarrow)$</u>:

$$\begin{aligned} S(n+1) = \text{ON} &\iff S(n) = \text{OFF} \\ &\iff n \equiv 0 \pmod 2 \qquad (\because P - I.H.) \\ &\iff n + 1 \equiv 1 \pmod 2 \end{aligned}$$

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Symbols & Words

- We first define a finite and non-empty set of **symbols** Σ.
- A **word** $w \in \Sigma^*$ is a sequence of symbols.
  - $\Sigma = \{0, 1\}$ – binary symbols.

$$\epsilon, 0, 1, 00, 01, 10010, \cdots \in \Sigma^*$$

  - $\Sigma = \{\mathsf{a}, \mathsf{b}, \cdots, \mathsf{z}\}$ – lowercase letters.

$$\epsilon, \mathsf{a}, \mathsf{b}, \mathsf{abc}, \mathsf{hello}, \mathsf{cs}, \mathsf{students}, \cdots \in \Sigma^*$$

  - $\Sigma = \{a \mid a \text{ is an Unicode character}\}$ – Unicode characters.

$$\epsilon, \text{안녕하세요}, \text{こんにちは}, \bigstar\blacksquare\blacktriangle\oplus, \cdots \in \Sigma^*$$

Symbols & Words

Notation	Description
ϵ	the **empty word**.
$w_1 w_2$	the **concatenation** of w_1 and w_2. (w_1 is a **prefix** of $w_1 w_2$ and w_2 is a **suffix** of $w_1 w_2$)
w^R	the **reverse** of w.
$\lvert w \rvert$	the **length** of w.
Σ^k	the set of all words of length k.
Σ^*	the set of all words (the **Kleene star**). (i.e., $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \cdots = \bigcup_{k \geq 0} \Sigma^k$)
Σ^+	the set of all words except ϵ (the **Kleene plus**). (i.e., $\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \cdots = \bigcup_{k \geq 1} \Sigma^k$)

Languages

A **language** $L \subseteq \Sigma^*$ is a specific set of words.

When $\Sigma = \{0, 1\}$, we can define the following languages:

- $L = \{\epsilon, 0, 1\}$ – the empty word, zero, and one.
- $L = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, \cdots\}$ – all binary words.
- $L = \{0^n 1^n \mid n \geq 0\}$ – equal number of consecutive zeros and ones.
- $L = \{10, 11, 101, 111, 1011, \cdots\}$ – ???

Languages – Operations

- The **union**, **intersection**, and **difference** of languages:

$$L_1 \cup L_2 \qquad L_1 \cap L_2 \qquad L_1 \setminus L_2$$

- The **reverse** of a language:

$$L^R = \{w^R \mid w \in L\}$$

- The **complement** of a language:

$$\overline{L} = \Sigma^* \setminus L$$

- The **concatenation** of languages:

$$L_1 L_2 = \{w_1 w_2 \mid w_1 \in L_1 \wedge w_2 \in L_2\}$$

Languages – Operations

- The **power** of a language defined inductively:

$$\begin{aligned} L^0 &= \{\epsilon\} \\ L^n &= L^{n-1}L \qquad (n \geq 1) \end{aligned}$$

- The **Kleene star** of a language:

$$L^* = L^0 \cup L^1 \cup L^2 \cup \cdots = \bigcup_{n \geq 0} L^n$$

- The **Kleene plus** of a language:

$$L^+ = L^1 \cup L^2 \cup L^3 \cup \cdots = \bigcup_{n \geq 1} L^n$$

- For any language L, are the following true?
  - $\epsilon \in L^*$ – Yes. Because $\epsilon \in L^0 \subseteq L^*$
  - $\epsilon \notin L^+$ – No. If $\epsilon \in L$, then $\epsilon \in L = L^1 \subseteq L^+$

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Next Lecture

- Basic Introduction of Scala

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